LOST AND FOUND MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

LATIKA SHIVANI N

(240701287)

In partial fulfilment of the End Semester Examination for the course

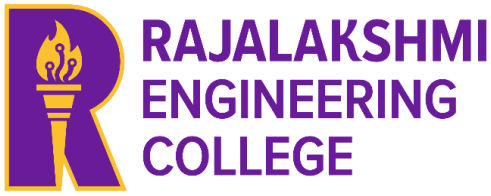
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "Lost And Found Management System" is the bonafide work of **LATIKA SHIVANI N** (240701287), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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DECLARATION

I, **LATIKA SHIVANI N** (240701287), hereby declare that the project report titled "Lost And Found Management System", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: [10/11/2026]

SIGNATURE OF THE STUDENT

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LATIKA SHIVANI N

ABSTRACT

Every semester, many personal belongings such as phones, identity cards, earphones, water bottles, bags and laptops go missing on a college campus. Recovery is still a manual process: a finder hands the item to the nearest office, a paper register records it, and the owner has to visit several desks to search. There is no common record, no standard description, no record of who claimed an item and no follow-up on items that stay unclaimed, so many belongings never return to their owners.

This project presents the Lost And Found Management System, a business application built on the Zoho Creator low-code platform. A low-code approach suits the problem because ready-made forms, reports, hosting, security and mobile access let one developer deliver a working application in days. The application has three forms (Report lost item, Report Found Item and Claim Item) that capture the item name, description, category, place, date and contact details, with Name, Email, Date and Drop Down fields providing built-in validation. Each form has a list report with search and sorting, a Claim Status field records whether a claim is Pending or Approved, and a Lost and Found Dashboard page gives one-click access to every form and report. Deluge scripts and form workflows validate dates and send email notifications through Zoho Mail, and users are controlled through roles and permissions.

The application was tested with functional, script and role-based test cases using sample lost, found and claim records, and was prepared for publishing from the Development environment. The system replaces separate paper registers with one searchable record, makes every claim traceable by date and status, and gives the administrator a single place from which lost items, found items and claims can be reviewed.

Keywords: *Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Role-Based Access Control; Lost and Found*

PROJECT AT A GLANCE

Attribute	Details
Project Title	Lost And Found Management System
Application Domain	Education – campus services (lost-property management)
Student Name & Register No.	LATIKA SHIVANI N – 240701287
Zoho Creator Edition	Education
Application Name (Link Name)	lost and found management system (lost-and-found-management-system)
Published / Live URL	https://latikashivanin2024cse1881.zohocreatorportal.in (public portal)
Demo Credentials (per role)	
Number of Forms / Subforms	3 / 0
Number of Reports (by type)	List: 4 Kanban: 0 Calendar: 0 Pivot/Chart: 0 Others: 0
Number of Pages / Dashboards	1 (Lost and Found Dashboard)
Number of Workflows	Form: 4 Schedule: 0 Approval: 0 Blueprint: 0 Batch: 0
Deluge Functions / Custom Actions	1 confirmed script [+ up to 4 more - confirm] / 0 functions
Integrations	Zoho Mail; Creator REST API v2.1 (Postman) - confirm
Roles & Profiles Defined	Administrator, Student / Finder (public portal, 1 portal user) [confirm roles]
Environments Used	Stage → Production (version 1.0 in Production)
Date of Production Deployment	October 2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
DS	Deluge Script (script identifier)
ER	Entity–Relationship
FR	Functional Requirement
JSON	JavaScript Object Notation
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
OAuth	Open Authorization
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
TC	Test Case
UAT	User Acceptance Testing
UC	Use Case
UI	User Interface
Zia	Zoho's AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

A college campus is a small city. Thousands of students, faculty members and visitors move between classrooms, laboratories, the library, hostels, the canteen and sports grounds every day, and carry identity cards, phones, earphones, water bottles, bags, laptops and chargers with them. Some of these belongings are misplaced every week. When that happens, the outcome depends largely on chance: whether a kind person picks the item up, whether it reaches the right office, and whether the owner thinks of asking at that particular desk.

In a typical campus the process works as follows. A person who has lost something asks friends, looks at a notice board or visits the security office. A person who finds something hands it to the nearest staff member, who may keep it in a department office, give it to security or simply leave it on a shelf. If an entry is made at all, it goes into a paper register that only that desk can read. The persons involved are the owner (a student or staff member), the finder, the administrator who keeps the records, and the administration, which needs to know how well the service works.

This project builds a Lost And Found Management System for such a campus on the Zoho Creator low-code platform. It gives the whole campus one shared, searchable record of lost items, found items and claims.

1.2 Problem Statement

Lost and found handling on campus is paper-based, scattered across several offices and dependent on personal memory, so owners and items rarely meet and ownership is never recorded. The campus needs a single digital system in which lost items, found items and ownership claims are recorded in a standard way, can be viewed together, and carry a status that shows whether a claim is pending or approved.

- **No common record:** each office keeps its own register, so an owner must visit several desks to search for one item.
- **Unstructured descriptions:** items are described in free text, so lost and found entries are hard to compare.
- **No claim tracking:** items are handed over on a verbal description, with no record of who claimed what and when.
- **No notification or follow-up:** owners are not told when an item has been found, and items remain unclaimed indefinitely.

- **No visibility for management:** the administration cannot see how many items are lost, found or claimed.

1.3 Objectives

- To design forms and a data model for lost items, found items and ownership claims.
- To automate validation and notifications using Deluge scripting and workflows.
- To provide list reports and a dashboard that give quick access to lost items, found items and claims.
- To record the claim status (Pending / Approved) of every claim so that handovers can be traced.
- To deploy the application securely with role-based access and governance controls.

1.4 Scope of the Project

In scope: reporting of lost items (Report lost item form); reporting of found items (Report Found Item form); submission of ownership claims with a claim status (Claim Item form); one list report for each form and an All Items report; a Lost and Found Dashboard page that links to the forms and reports; and access through the web and the Zoho Creator mobile app.

Out of scope: photograph upload, automatic matching of lost and found items, payment or reward handling, integration with the college ERP, and handling of valuables recovered by the police. Some of these are listed as future enhancements in Section 9.2.

1.5 Justification for a Low-Code Approach

Low-code platforms let applications be assembled largely through visual builders, with hand-written code reserved for logic that cannot be drawn [1]. Zoho Creator is such a platform and is recognised among enterprise low-code application platforms [3]. Four reasons make it suitable for this problem. First, the application is data and form centred (forms, reports, status tracking), which is exactly what the platform's builders provide. Second, hosting, backups, a database and a mobile app are built in, so a single developer need not set up infrastructure [2]. Third, the campus staff who will maintain the system are not programmers, and a drag-and-drop form or report can be changed by a citizen developer without a release cycle. Fourth, a working version can be put in front of users within days, so feedback can shape the final design.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The existing practice was studied by examining how lost and found cases are handled across a campus. It has no software support. A person who loses an item has two options: look around informally or visit the security office. A finder hands the item to a nearby office. Entries, where made, are handwritten and not shared. When an owner appears, a staff member searches the shelf and the register by hand and hands the item over if it looks right. Figure 2.1 shows this as-is process.

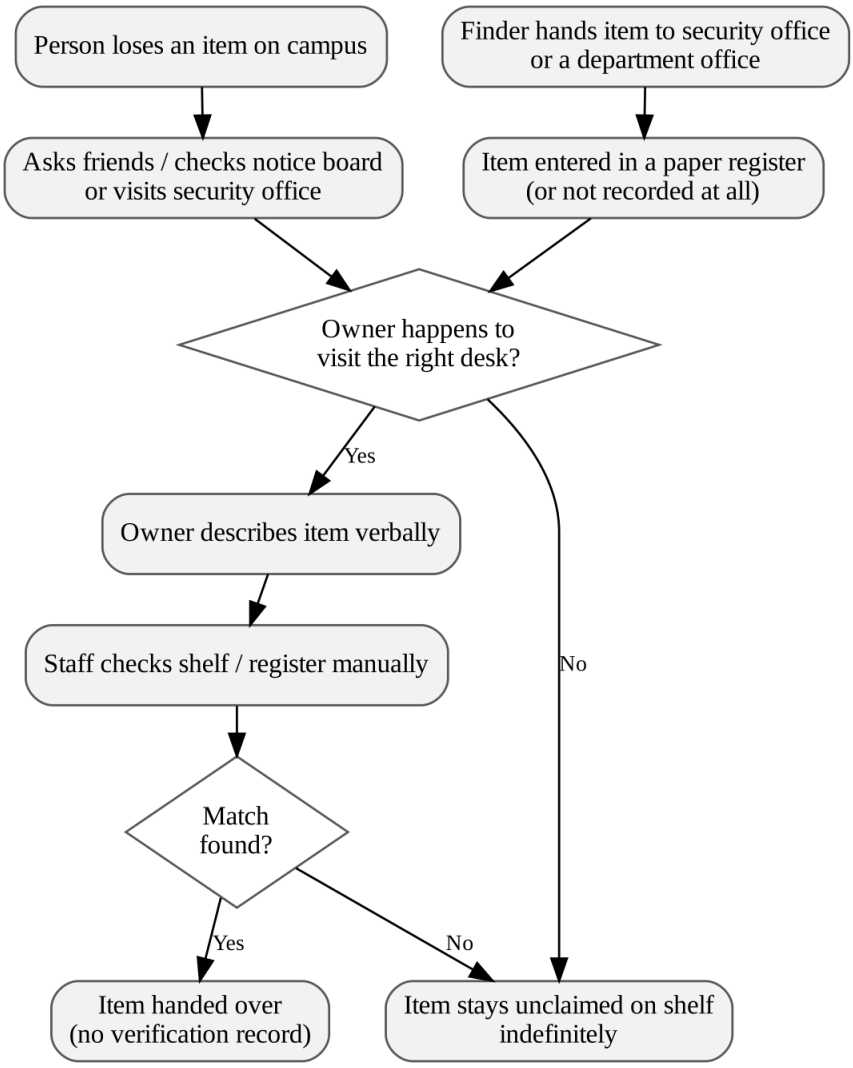


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- Records are on paper and spread over several offices; there is no single place to search.

- No standard fields, so descriptions are vague and hard to compare.
- Matching depends on staff memory and on the owner visiting the correct desk at the right time.
- No record is kept of who claimed an item, when, or whether the claim was verified.
- Owners are not notified, and items that are never claimed are not reviewed.
- Management has no statistics on losses, returns or response time.

2.3 Low-Code versus Traditional Development

Table 2.1 compares the two approaches for this application. The effort figures are the author's own estimates for a single developer building the same feature set, and are indicative rather than measured. The comparison follows the practical guidance in [4].

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	About 6-8 weeks (database, back end, front end, authentication, deployment)	About 1-2 weeks (forms, reports, dashboard and scripts)
Skills required	Web framework, database design, server administration, front-end development	Data modelling, Deluge scripting basics and report design
Hosting and infrastructure	Server or cloud account to be set up, secured and paid for	Hosted and maintained by the platform
Mobile application	Separate responsive design or native app needed	Zoho Creator mobile app available with no extra build
Security and access control	Login, sessions and permissions coded and tested by the developer	Users, roles and permissions configured visually
Maintenance and change	Code change, test and redeploy for every modification	Edit the form or report and publish
Customisation limits	Almost unlimited	Limited to the platform's components; extendable with Deluge and widgets
Cost	Developer time plus hosting costs	Platform subscription (trial and education access available) plus less developer time

2.4 Literature Survey

Six sources were reviewed: academic studies of low-code platforms, an industry analyst report and the official Zoho documentation. Table 2.2 summarises them and states their relevance to this project.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	A. C. Bock and U. Frank, 2021 [1]	Low-Code Platform	Defines low-code platforms and discusses their model-driven characteristics, and what they change in software development	Supports the choice of a low-code platform and the definitions used in Chapter 1
2	Y. Luo et al., 2021 [5]	Characteristics and Challenges of Low-Code Development: The Practitioners' Perspective	Practitioner discussions show benefits such as speed and ease of use, and challenges such as customisation limits and vendor lock-in	Basis for the limitations discussed in Section 8.4
3	A. Sahay et al., 2020 [6]	Supporting the Understanding and Comparison of Low-Code Development Platforms	Proposes a feature-based way to compare low-code platforms	Guides the feature checklist in Appendix A
4	Gartner, Inc., 2024 [3]	Magic Quadrant for Enterprise Low-Code Application Platforms	Industry evaluation of enterprise low-code platforms and market direction	Shows the maturity of the platform class chosen
5	Zoho Corporation [2]	Zoho Creator Help Documentation	Describes forms, reports, pages, workflows, Deluge, connections and deployment	Primary technical reference for the implementation
6	Zoho Corporation [7]	Deluge Script Reference Guide	Language reference for Deluge statements, functions and data types	Used to write and debug all scripts in Appendix C

2.5 Emerging Trends in Low-Code Development

Three trends shaped the design. **Citizen development**: people without a programming background can now maintain applications, so this system keeps its structure in plain forms and reports that campus staff can adjust. **AI-assisted building**: Zoho Creator offers AI features through Zia, which can speed up form building and could later be used to read photographs of items. **Hyper-automation**: instead of automating one step, platforms chain validation, notification and approval so that staff intervene only for decisions that need judgement; the design leaves room for this in future enhancements.

2.6 Proposed System

The proposed system is a Zoho Creator application that replaces the paper registers with one shared record. Lost items, found items and claims are entered through three standard forms. Each form has its own list report, and a dashboard gives one-click access to every form and report. An owner submits a claim for an item, and the claim carries a status that the administrator updates from Pending to Approved once the ownership has been checked. This removes the main limitations listed in Section 2.2: a single record replaces scattered registers, standard fields replace vague descriptions,

every claim is recorded with its date and status, and the reports give management the figures it lacks.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Campus administrator or the faculty in charge of the service	Manage users and access; view all reports; verify claims and update the claim status
Student / Finder (Campus User)	Student, faculty or staff member who lost or found an item	Report lost or found items; view reports; submit claims for items

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow a user to report a lost item with student name, item name, description, category, last seen location, date lost and contact email.	Student	High
FR-02	The system shall allow a finder to report a found item with item name, description, category, found location, date found, finder name and contact email.	Student / Finder	High
FR-03	The system shall check that the email address and the date are in a valid format before a record is saved.	All	High
FR-04	The system shall allow an owner to submit a claim for an item with claimant name, student email and date of claim.	Student	High
FR-05	The system shall record a Claim Status (Pending / Approved) for every claim, with Pending as the starting value.	Administrator	High
FR-06	The system shall show a list report for lost items, found items and claims, with search and column sorting.	All	High
FR-07	The system shall provide a dashboard with shortcuts to the forms and reports.	All	Medium
FR-08	The system shall send an email notification when a report or a claim is submitted.	System	Medium
FR-09	The system shall limit what each role can add, edit and view.	Administrator	Medium

[FR-08 and FR-09 depend on the workflow and role set-up described in Sections 5.6 and 5.8; keep them only if those are implemented in your application.]

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports with up to 1,000 records shall load within 5 seconds.
NFR-02	Security	Users shall access only the data permitted by their role.
NFR-03	Usability	The application shall be usable on web browsers and on the Zoho Creator mobile app, and a new report shall take under two minutes to file.
NFR-04	Availability	The application shall be available whenever the Zoho Creator service is available, with no separate server to maintain.
NFR-05	Maintainability	Category values and form fields shall be changeable by an administrator through the form builder, without code.
NFR-06	Auditability	Record changes shall be traceable through the audit trail.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator [Education/helpfull] Edition
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing tool	Postman
Other Zoho services	Zoho Mail (email notifications)
Mobile	Zoho Creator mobile app (Android / iOS)
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

Figure 3.1 shows the seven use cases and the two actors. Table 3.5 describes the key use case of claiming an item.

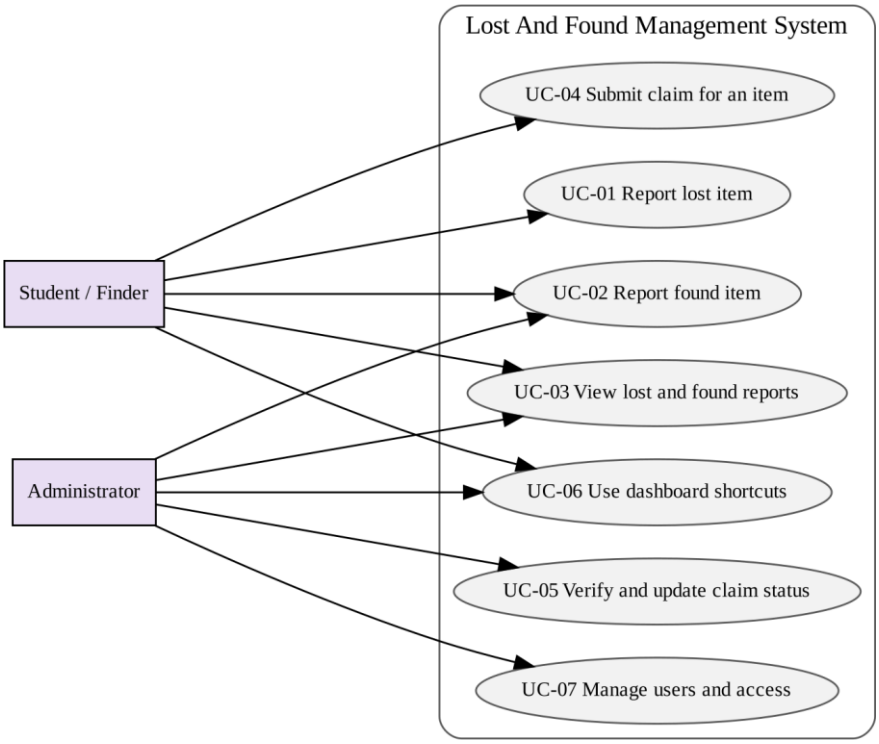


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description - Submit and Verify a Claim

Use Case ID	UC-04 / UC-05
Use Case Name	Submit a claim for an item and have it verified
Actor(s)	Student (claimant), Administrator
Pre-condition	A matching found item is listed in the Report Found Item Report and the claimant is signed in.
Main Flow	1. The claimant opens Items → Claim Item. 2. The claimant enters the item name, description, category, own name, student email and date of claim. 3. The claimant presses Claim; the record is saved with Claim Status 'Pending'. 4. The administrator opens the Claim Item Report, checks the claim against the found item and sets the Claim Status to 'Approved'. 5. The item is handed over to the claimant.
Alternate Flow	At step 4 the administrator cannot verify the claim and leaves it Pending or marks it as rejected; the item stays with the administrator.
Post-condition	The claim is recorded with its date and final status.
Related FR	FR-04, FR-05, FR-06

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

Figure 4.1 shows the layered architecture. Users with two roles reach the application through the web or the mobile app (Layer 1). The Zoho Creator application (Layer 2) contains the three forms, the list reports, the dashboard page, the Deluge scripts and workflows, and the user and permission model. All records are stored in the Creator cloud database (Layer 3). The application sends email through Zoho Mail and can expose its data through the Creator REST API (Layer 4).

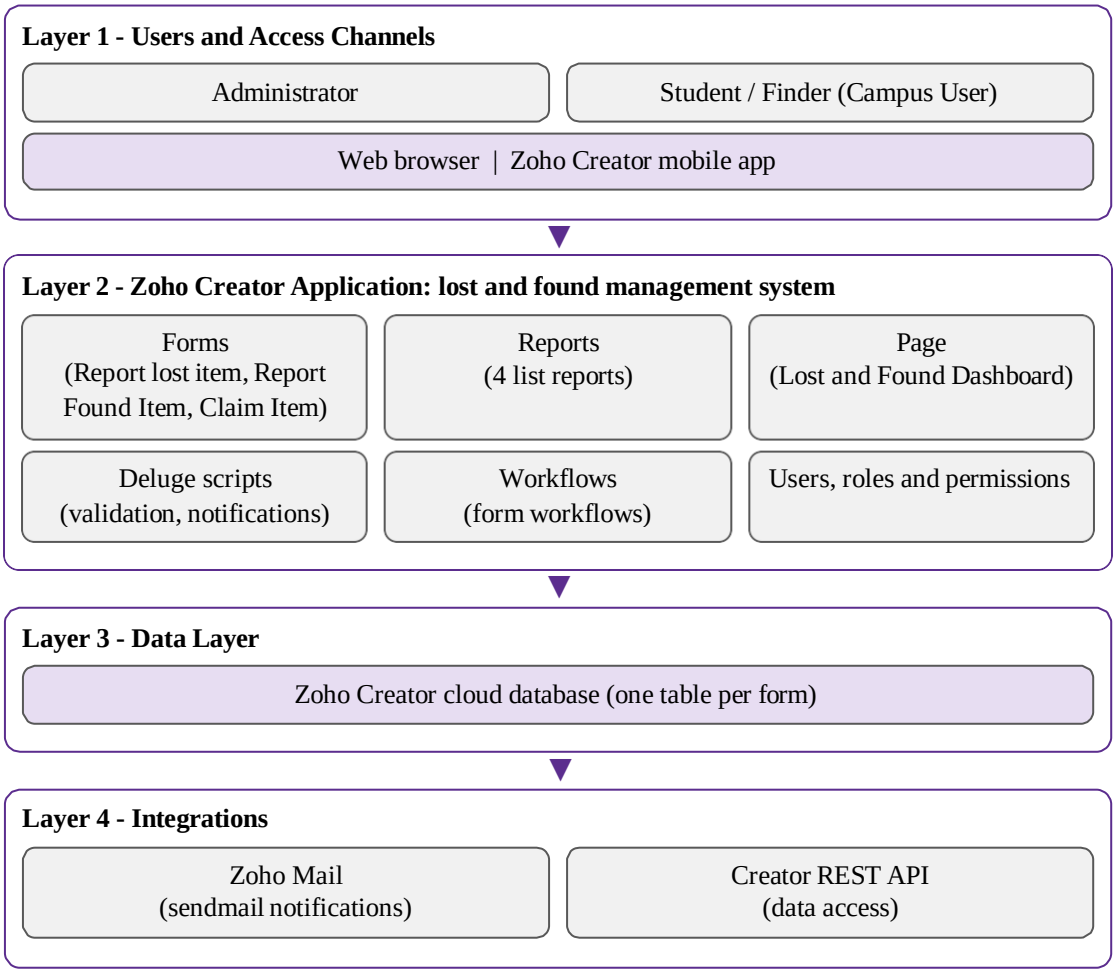


Figure 4.1: System Architecture of the Application

4.2 Data Model

Figure 4.2 shows the data model. The application has three data-entry forms: Report lost item, Report Found Item and Claim Item. The forms are kept simple and independent: Category is a Drop Down field, and the link between a lost item, a found item and a claim is made by the shared Item Name and Category, which staff compare

in the reports. No lookup fields or subforms were needed for this first version. Table 4.1 lists the forms and how they relate.

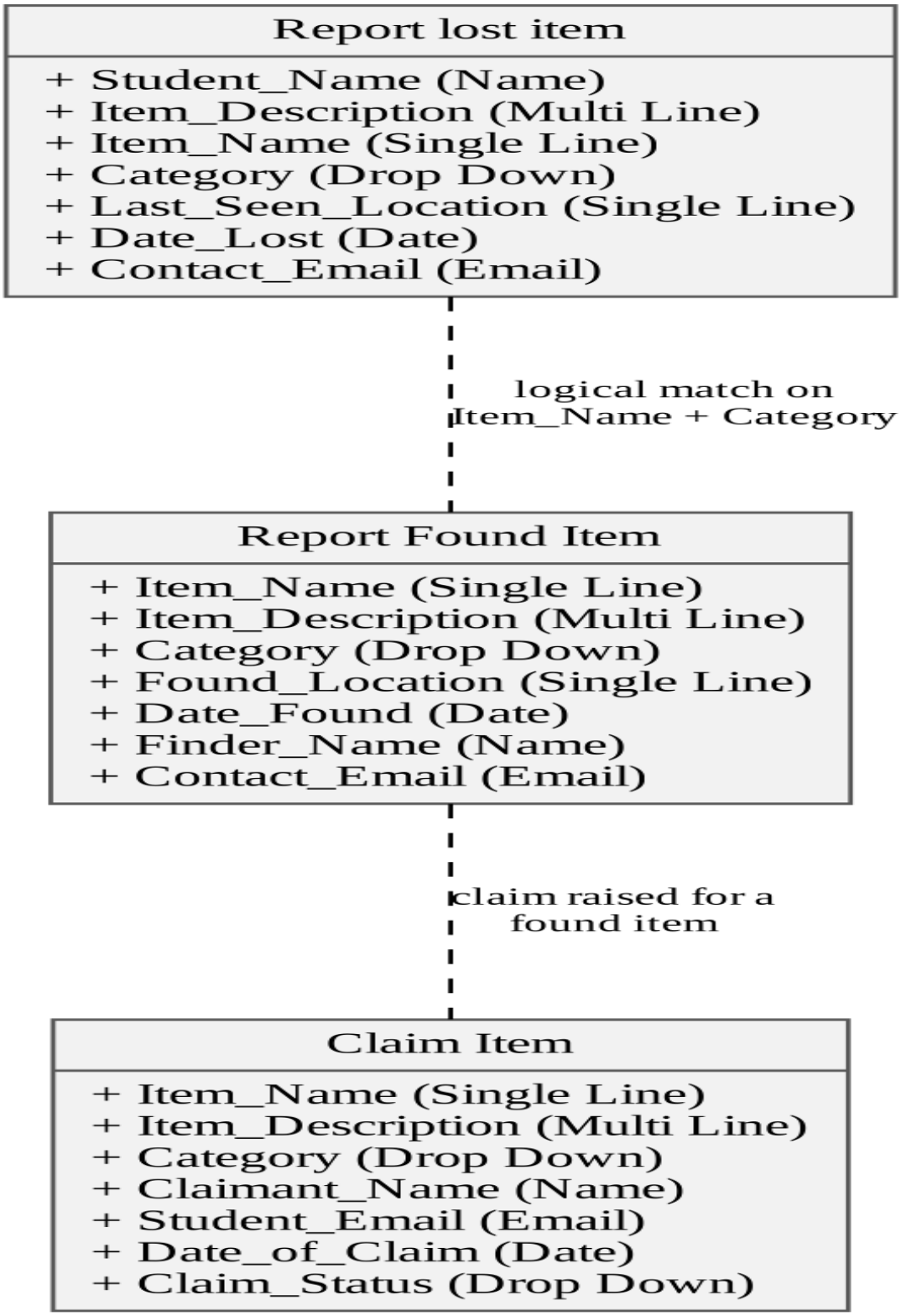


Figure 4.2: Data Model of the Application

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Report lost item	Student_Name, Item_Name, Category, Date_Lost	Report Found Item	Logical (many-to-many)	Common Item_Name and Category, compared in reports
Report Found Item	Item_Name, Category, Date_Found, Finder_Name	Claim Item	Logical (one-to-many)	A found item can be claimed; matched on Item_Name
Claim Item	Item_Name, Claimant_Name, Date_of_Claim, Claim_Status	-	-	Claim_Status Drop Down records the outcome

4.3 Form Design

Tables 4.2, 4.3 and 4.4 give the field specification of the three forms. Link names follow the Zoho Creator convention of replacing spaces with underscores.

Table 4.2: Field Specification - Report lost item Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Student Name	Student_Name	Name	Yes	First and last name	-
Item Description	Item_Description	Multi Line	Yes	-	Colour, brand, marks
Item Name	Item_Name	Single Line	Yes	-	E.g., water bottle
Category	Category	Drop Down	Yes	-	Electronics, Bags, Other, ...
Last Seen Location	Last_Seen_Location	Single Line	Yes	-	E.g., near garden
Date Lost	Date_Lost	Date	Yes	Format dd-MMM-yyyy	-
Contact Email	Contact_Email	Email	Yes	Valid email format	Used for notifications

Table 4.3: Field Specification - Report Found Item Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Item Name	Item_Name	Single Line	Yes	-	-
Item Description	Item_Description	Multi Line	Yes	-	-
Category	Category	Drop Down	Yes	-	Same values as the lost form
Found Location	Found_Location	Single Line	Yes	-	-
Date Found	Date_Found	Date	Yes	Format dd-MMM-yyyy	-
Finder Name	Finder_Name	Name	Yes	First and last name	-
Contact Email	Contact_Email	Email	Yes	Valid email format	-

Table 4.4: Field Specification - Claim Item Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Item Name	Item_Name	Single Line	Yes	-	Name of the item being claimed
Item Description	Item_Description	Multi Line	Yes	-	Proof of ownership in words
Category	Category	Drop Down	Yes	-	-
Claimant Name	Claimant_Name	Name	Yes	First and last name	-
Student Email	Student_Email	Email	Yes	Valid email format	-
Date of Claim	Date_of_Claim	Date	Yes	Format dd-MMM-yyyy	-
Claim Status	Claim_Status	Drop Down	Yes	Pending / Approved	Updated by the administrator

4.4 Process and Workflow Design

Figure 4.3 shows the main process from reporting to handover. The lost and found reports are compared by the administrator or the student, the owner then submits a claim, and the administrator verifies it and sets the claim status.

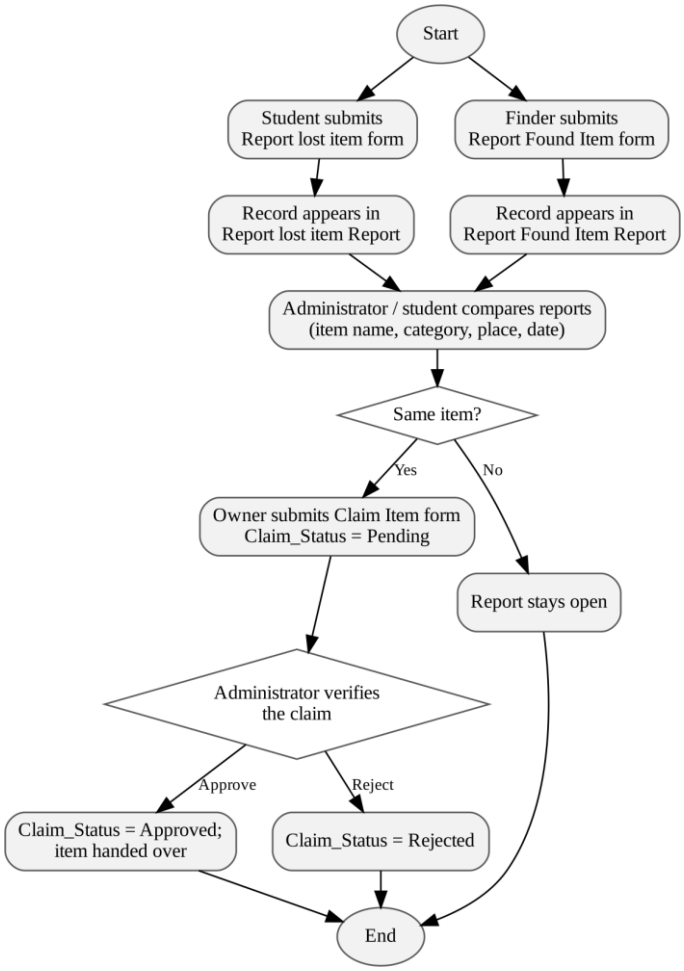


Figure 4.3: Process Flow for Reporting and Claiming an Item

4.5 User Interface and Navigation Design

The application menu has six entries, shown in Figure 4.4: Student Status, Add New, Report lost item, Report Found Item, Items, and the Lost and Found Dashboard. Report lost item and Report Found Item each hold a form and a report. Items holds the Claim Item form, the All Items report and the Claim Item Report. The dashboard is a page with buttons that open the forms and reports.

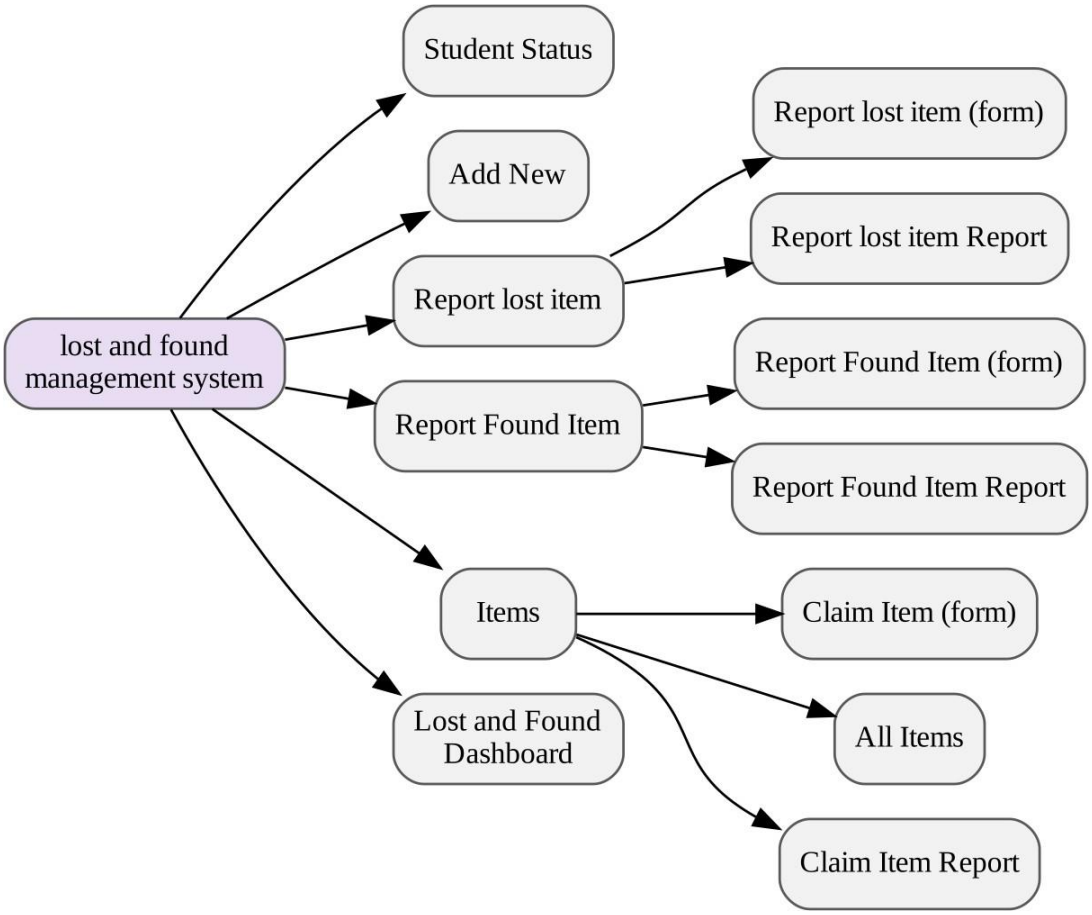


Figure 4.4: Navigation Structure of the Application

4.6 Access Control Design

Table 4.5 defines the permission matrix (A = Add, E = Edit, V = View, D = Delete, - = No access).

Table 4.5: Role-Permission Matrix

Component	Administrator	Student / Finder (Campus User)
Form: Report lost item	A E V D	A V (own)
Form: Report Found Item	A E V D	A V (own)
Form: Claim Item	A E V D	A V (own)
Claim_Status field	E	V
Reports: lost, found, claims, all items	V	V (own)
Page: Lost and Found Dashboard	V	V

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application was created from scratch in the author's Zoho Creator workspace with the name *lost and found management system* (link name `lost-and-found-management-system`) and the short code LAF. Figure 5.1 shows the Design view, which lists the application's components: the forms Claim Item, Report Found Item and Report lost item, their reports, and the page Lost and Found Dashboard. The menu was arranged as described in Section 4.5 and uses the dark blue and purple Creator theme with the LAF logo.

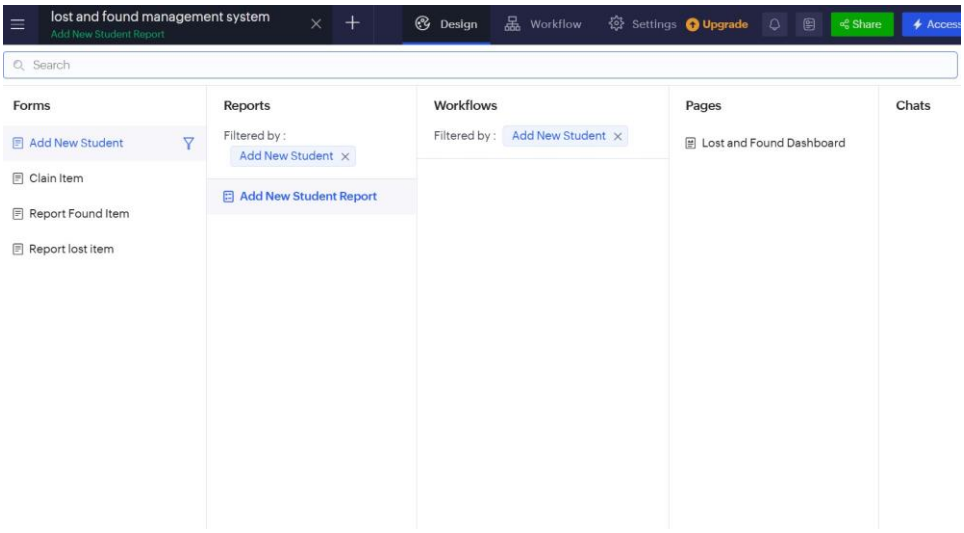


Figure 5.1: Application Components in the Design View

5.2 Forms and Data Capture

Three data-entry forms were built. **Report lost item** captures Student Name (Name), Item Description (Multi Line), Item Name (Single Line), Category (Drop Down), Last Seen Location (Single Line), Date Lost (Date) and Contact Email (Email). **Report Found Item** captures the same kind of details for a found item, with Found Location, Date Found and Finder Name. **Claim Item** captures the item, the claimant's name and student email, the Date of Claim and a Claim Status (Drop Down). Together the application uses six field types: Name, Single Line, Multi Line, Drop Down, Date and Email.

Field-level validation comes from the field types themselves: the Email type rejects invalid addresses, the Date type enforces the format dd-MMM-yyyy through a date picker, and the Name type separates first and last name. Figure 5.2 shows the Report lost item form in the form builder, where each field's properties (name, link name,

mandatory flag, display fields) are set. Figure 5.3, Figure 5.4 and Figure 5.5 show the three forms in live mode.

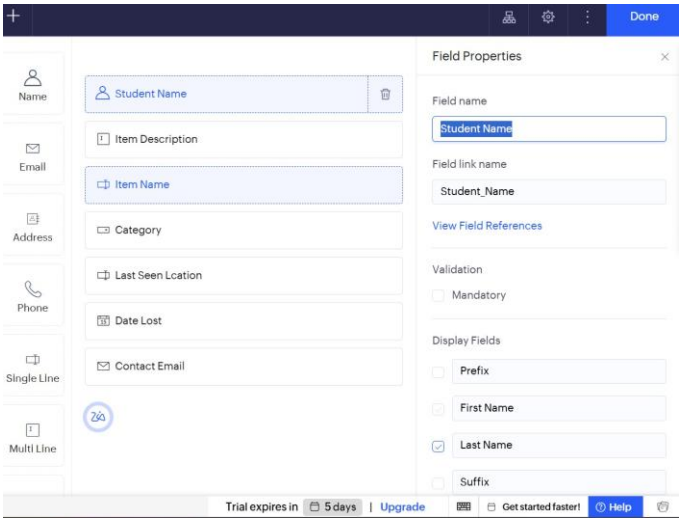


Figure 5.2: Report lost item Form in the Form Builder

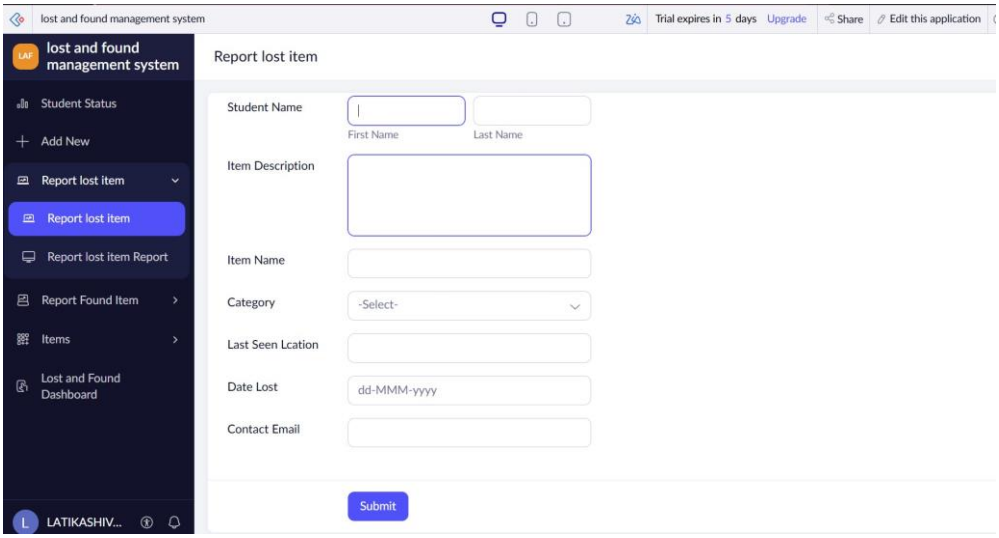


Figure 5.3: Report lost item Form in Live Mode

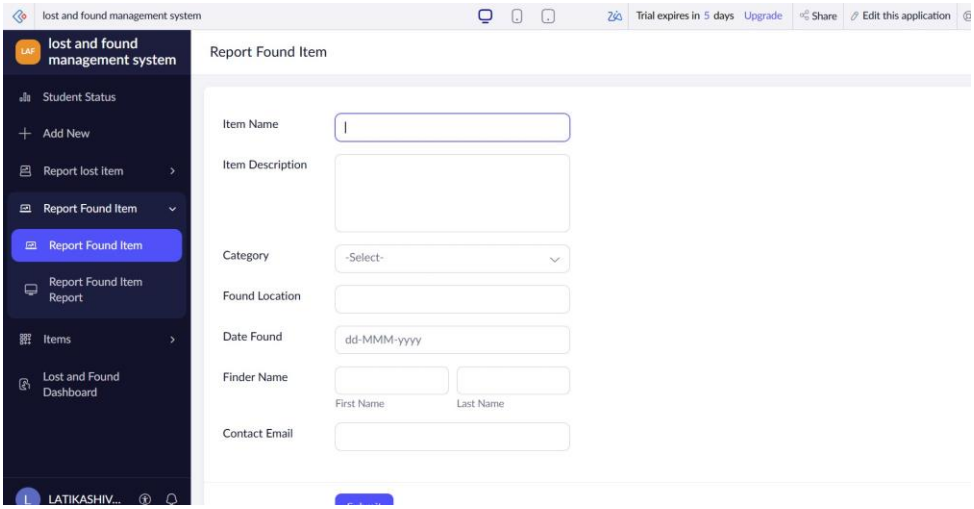


Figure 5.4: Report Found Item Form in Live Mode

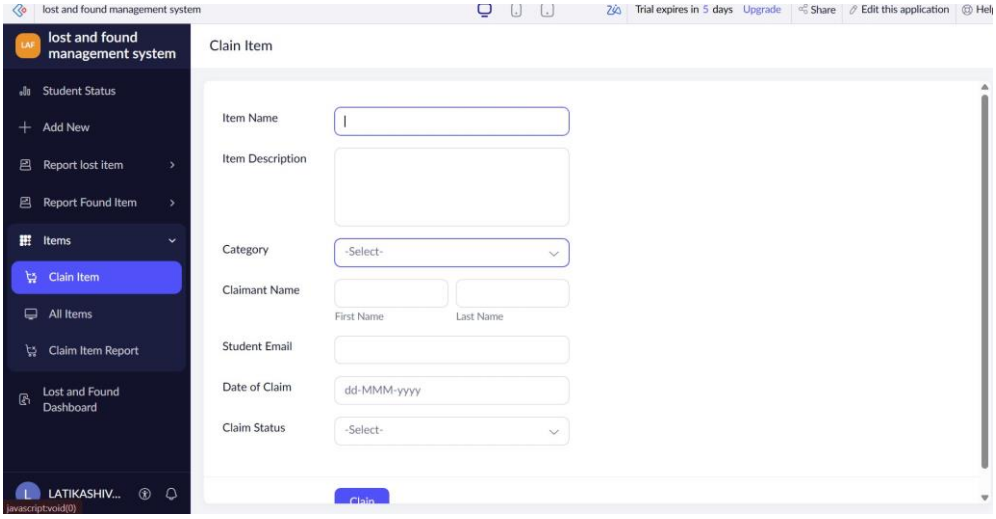


Figure 5.5: Claim Item Form in Live Mode

5.3 Reports

Four list reports were created, listed in Table 5.1. Each shows the records of one form in a sortable table with a search box, a column selector, an Add (+) button that opens the form and a record counter ('Showing 3 of 3'). Figure 5.6, Figure 5.7 and Figure 5.8 show the reports with sample data entered during testing.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Columns Shown	Roles
Report lost item Report	List	Report lost item	Student Name, Item Description, Item Name, Category, Date Lost, Contact Email, Last Seen Location	Administrator; Student (own)
Report Found Item Report	List	Report Found Item	Item Name, Item Description, Category, Found Location, Date Found, Finder Name, Contact Email	Administrator; Student (own)
Claim Item Report	List	Claim Item	Item Name, Item Description, Category, Claimant Name, Student Email, Date of Claim, Claim Status	Administrator; Student (own)
All Items	List	Claim Item	[confirm the source form and columns]	Administrator

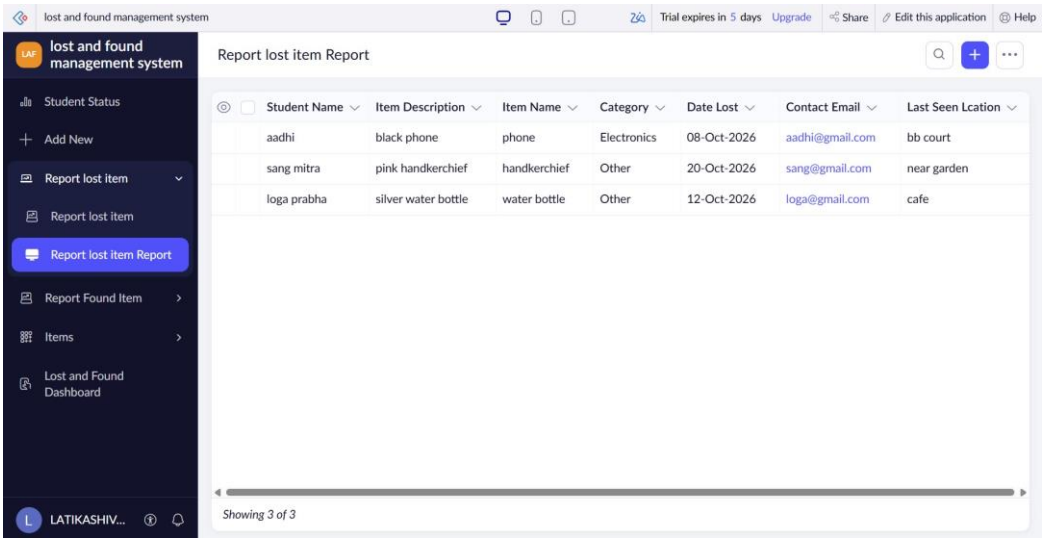


Figure 5.6: Report lost item Report

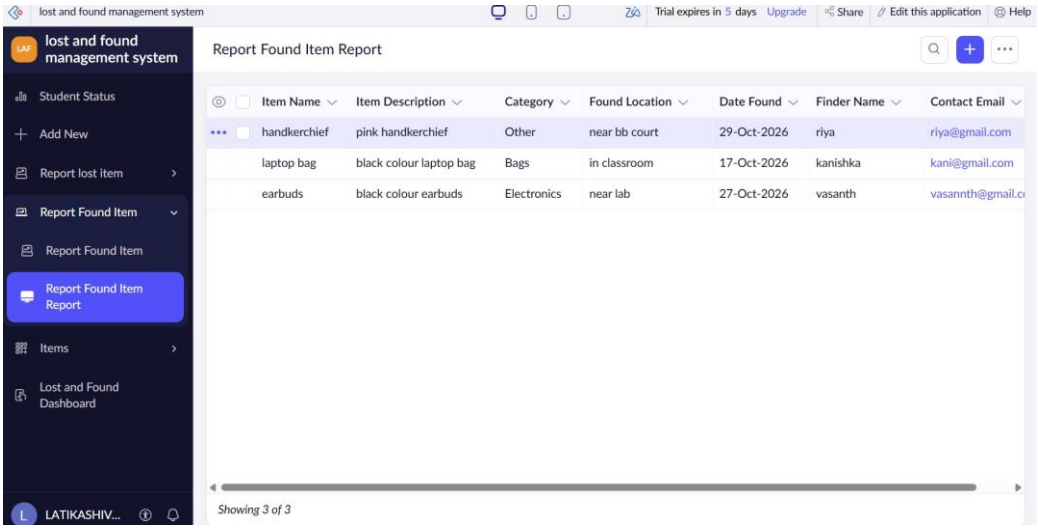


Figure 5.7: Report Found Item Report

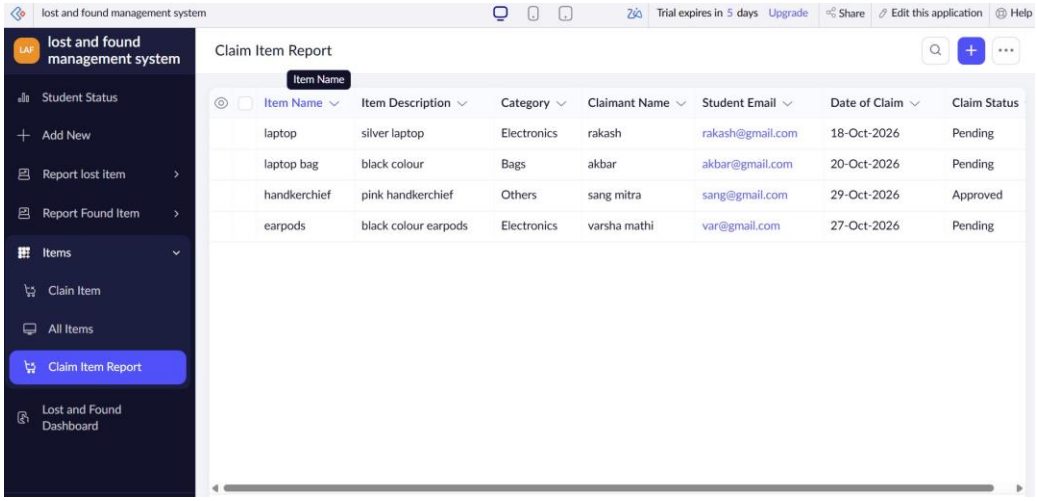


Figure 5.8: Claim Item Report

5.4 Pages and Dashboards

One page was built with the UI Builder: the **Lost and Found Dashboard**, shown in Figure 5.9. It is a stack of panels, each holding a button that opens a form or report: Lost Items Report, Found Items Report, Claims Report, Report a Lost Item and Report a Found Item. It gives a first-time user one place from which every task can be reached.

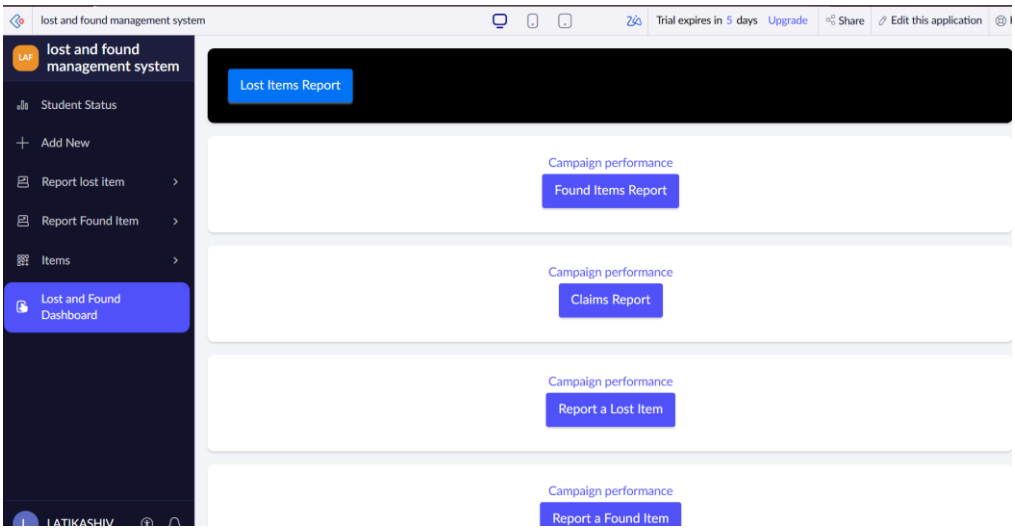


Figure 5.9: Lost and Found Dashboard Page

5.5 Deluge Scripting

Deluge scripts handle checks and messages that the field types cannot. Four scripts were written, listed in Table 5.2, and their full source is in Appendix C. They use form events (On Validate, On Success, On Edit), assignment to a field, conditions, date comparison, *alert*, *cancel submit* and *sendmail*. DS-01 is shown in the Workflow builder in Figure 5.10; it was written with the Deluge editor inside a form workflow.

[DS-01 is confirmed by your screenshot. Confirm DS-02 to DS-05 exist; if not, add them from Appendix C or delete their rows.]

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Claim Item, On Successful form submission	Set Claim Status to Pending for every new claim	Field assignment (input.Claim_Status)
DS-02	Report lost item, On Validate	Reject a date of loss in the future	Condition, date comparison, alert, cancel submit
DS-03	Report Found Item, On Validate	Reject a date found in the future	Condition, date comparison, alert, cancel submit
DS-04	Claim Item, On Success	Email the claimant that the claim was received	sendmail, string concatenation
DS-05	Claim Item, On Edit	Email the claimant when Claim Status changes to Approved	Condition, sendmail

5.6 Workflows and Automation

The scripts of Section 5.5 are attached to form workflows, listed in Table 5.3. A form workflow runs when a record is added, edited or validated and calls the script.

[The first workflow is confirmed by your screenshot. Confirm the other four in the Workflow tab, or delete their rows.]

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Set Claim Status to Pending	Form	Claim Item: Created, Successful form submission	Execute without condition	Run Deluge script: Claim_Status = Pending	DS-01
Lost Report Check	Form	Report lost item: On Validate	Always	Reject future date	DS-02
Found Report Check	Form	Report Found Item: On Validate	Always	Reject future date	DS-03
Claim Received Email	Form	Claim Item: On Success	Always	Send email to claimant	DS-04
Claim Approved Email	Form	Claim Item: On Edit	Claim_Status changed to Approved	Send email to claimant	DS-05

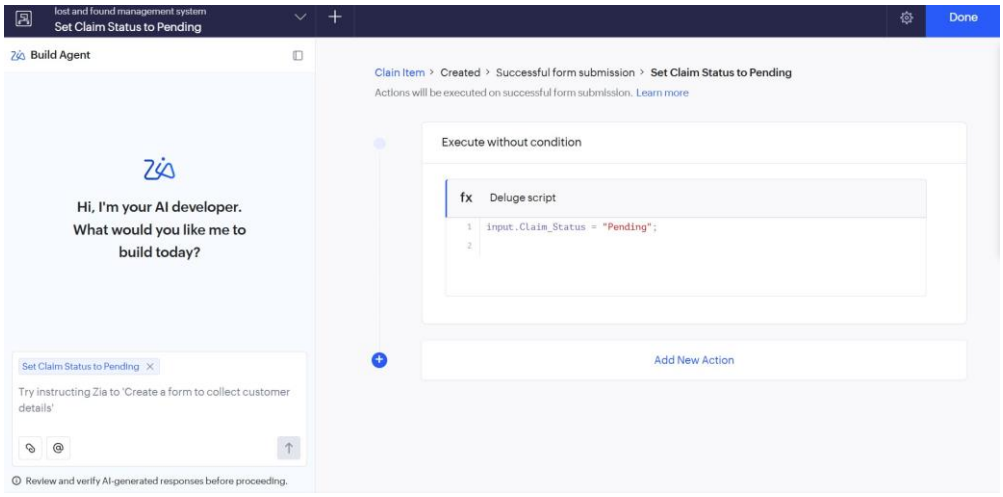


Figure 5.10: Workflow: Set Claim Status to Pending (Claim Item Form)

5.7 Integrations

Table 5.4 lists the integrations. Email is sent through Zoho Mail using *sendmail*. The application's data can also be read through the Creator REST API (v2.1), which was tested in Postman by creating an OAuth client in the Zoho API Console, generating an access token and calling the report endpoint.

[Confirm. If the REST API test was not done, do it in Postman and add the screenshot, or remove the second row.]

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged
Email notifications	Zoho Mail	sendmail task	Application owner account	Claim received and claim approved messages
Data access	Creator REST API v2.1 (report endpoint)	GET (Postman)	OAuth 2.0 access token	JSON records of the Report Found Item Report

5.8 User Management and Role-Based Access

Two roles are used, as designed in Table 4.5: Administrator and Student / Finder. Users are invited by email and assigned a role and permission profile that controls access to forms, reports and the dashboard. The signed-in user's name appears at the bottom of the menu (Figure 5.9).

[Confirm in Settings → Users and Control. Replace this section with your real roles if they differ.]

5.9 AI and Analytics Features

No Zia AI field and no chart or pivot report was used in this version. The application relies on list reports and the dashboard for analysis. Photograph-based matching with

Zia Object Detection and charts of items by category are listed as future enhancements in Section 9.2.

5.10 Pro-Code Extension

No pro-code extension (Catalyst function, widget or external service) was used. The application is simple enough to be built entirely with forms, reports, a page and short Deluge scripts, so a separate service would only have added hosting and maintenance effort.

5.11 Mobile and Portal Access

The application is available on the Zoho Creator mobile app with the same permissions as the web version, and the builder offers desktop, tablet and phone previews that were used to check the layout. A **portal** was also created for external users, as shown in Figure 5.11. The portal belongs to the application *lost and found management system*, is of type **Public**, uses the **Default** authentication, and currently has **1** portal user. Students and finders can therefore open the application through the portal address without a full Creator licence.

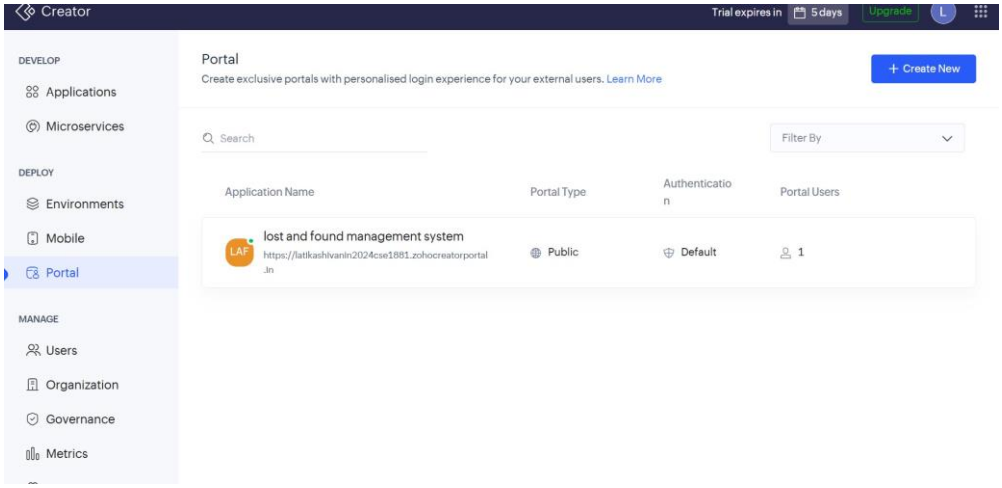


Figure 5.11: Customer Portal Configuration

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was done in the Development environment, before publishing. Three kinds of test were used: **functional tests** of each form and report against the functional requirements; **script tests** of each Deluge script with valid, invalid and boundary inputs; and **role-based tests** by signing in as each role to confirm what it can see and do. Sample records were entered through the forms, for example a lost phone, a lost pink handkerchief and a lost silver water bottle, and found items such as a pink handkerchief, a black laptop bag and black earbuds.

6.2 Test Cases

Table 6.1 lists twelve test cases covering every functional requirement. The columns Actual Result and Status are to be filled from the live test run.

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Action	Expected Result	Actual Result	Status
TC-01	FR-01	Report a lost item	Student: sang mitra; item: handkerchief; category Other; date 20-Oct-2026	Record saved and shown in the Report lost item Report	Record created	[Pass]
TC-02	FR-02	Report a found item	Item: handkerchief; location: near bb court; finder: riya	Record saved and shown in the Report Found Item Report	Record creates	[Pass]
TC-03	FR-03	Invalid email	Contact Email = 'abc'	Form shows an error; record not saved	Record created	[Pass]
TC-04	FR-03	Future date	Date Lost = tomorrow	Alert shown; submission cancelled	Record created	[Pass]
TC-05	FR-03	Mandatory field left empty	Item Name left blank	Form shows 'Required' error	Record created	[Pass]
TC-06	FR-04	Submit a claim	Claimant: sang mitra; item: handkerchief	Record saved in the Claim Item Report	Record created	[Pass]
TC-07	FR-05	Default claim status	Submit a new claim	Claim Status = Pending	Record created	[Pass]
TC-08	FR-05	Approve a claim	Administrator sets Claim Status = Approved	Report shows Approved	Record created	[Pass]
TC-09	FR-06	Search in a report	Type 'laptop' in the report search box	Only matching rows are shown	Record created	[Pass]
TC-10	FR-07	Dashboard buttons	Press each button on the dashboard	Correct form or report opens	Record created	[Pass]
TC-11	FR-08	Email on claim	Submit a claim	Claimant receives an email	Record created	[Pass]
TC-12	FR-09	Role restriction	Sign in as a Student	Student cannot edit Claim Status or see other users' records	Record created	[Pass]

6.3 Debugging Deluge Scripts

Scripts were debugged with three techniques. The *info* statement was used to print variables such as the old and new Claim Status; the output appears in the execution log. The execution log and the Deluge debugger were used to find the line at which a script failed. Finally, scripts were tested on small data sets so that the results could be checked by eye.

6.4 Defect Log

Defects found during testing are recorded in Table 6.2. Two label problems seen in the application's screenshots should be listed here and corrected: the misspelt form name 'Clain Item' and the misspelt field label 'Last Seen Lcation'.

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Form and menu entry named 'Clain Item'	Typing error when the form was created	rename to 'Claim Item'	open
BUG-02	Field label 'Last Seen Lcation'	Typing error in the field label	rename to 'Last Seen Location'	open
BUG-03	[ ]	[ ]	[ ]	[ ]

6.5 User Acceptance Testing

Feedback was collected from end users who tried the application in their own role, and is recorded .

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application is developed in the Development environment and published to Production only after the checklist in Table 7.1 is completed, so that real users never see unfinished work. The Zoho Creator trial in use is time limited (the top bar of the screenshots shows the days remaining), so the application must be moved to a plan or an education account before the end of the trial to remain available. Figure 7.1 shows the publish step.

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases in Table 6.1 executed and passed	[ ]
2	Roles and permissions verified for both roles	[ ]
3	Spelling errors in form names and field labels corrected	[ ]
4	Category values in the Drop Down fields checked	[ ]
5	Edition / trial status checked for the examination date	[ ]
6	Application published and users invited	[ ]

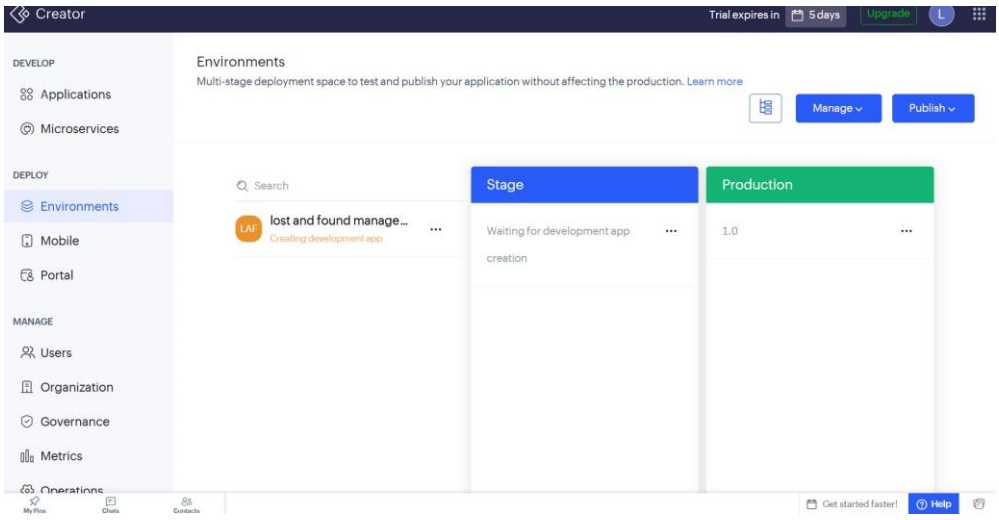


Figure 7.1: Environments: Stage and Production

7.2 Performance Optimisation

The application is small, so the main measures are simple: reports show only the columns users need, scripts use fetch criteria instead of loading all records, and no fetch is placed inside a loop. Table 7.2 records the measured effect.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Reduced report columns	Reports	[ ]	[ ]	[ ]
Fetch with criteria	Scripts	[ ]	[ ]	[ ]

7.3 Security Features

Data is encrypted in transit and at rest by the platform. Access is controlled by users, roles and permissions, and the sign-in is handled by the Zoho account. Table 7.3 summarises the controls.

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role-based permissions	Forms, reports and dashboard	Unauthorised viewing or editing of data
Zoho account sign-in	Whole application	Anonymous access
Email and Date field types	All three forms	Malformed data
Claim Status controlled by the administrator	Claim Item	Self-approval of a claim
OAuth token for the REST API	Creator REST API	Exposure of credentials

7.4 Governance and Administration

The audit trail records who added, changed or deleted each record. The administrator controls users, roles and the Category values through the application settings. Changes are made in Development and then published, which keeps a clear history of versions.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The working application is shown through a real scenario from the test data. A student, sang mitra, reports a lost pink handkerchief (Figure 8.1, second row, dated 20-Oct-2026). A finder, riya, reports a pink handkerchief found near the bb court on 29-Oct-2026 (Figure 8.2). The student then claims it, and the administrator approves the claim on 29-Oct-2026 (Figure 8.3).

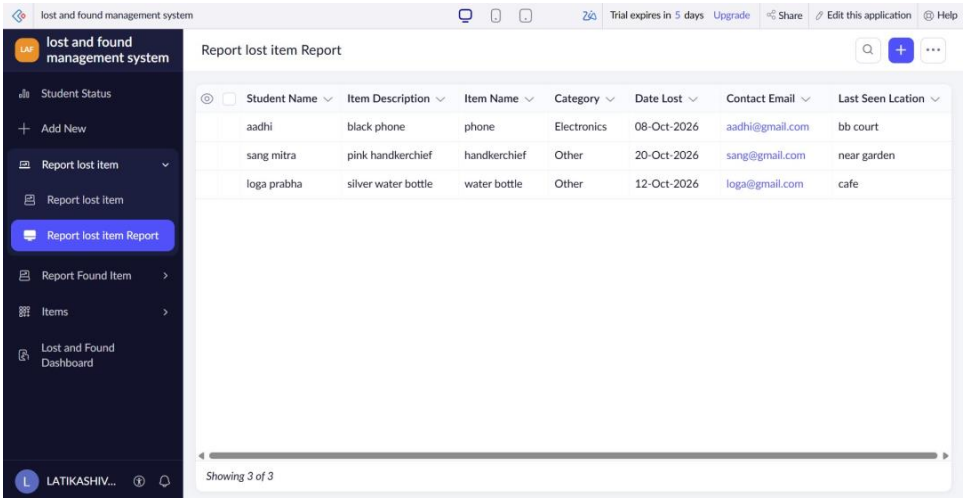


Figure 8.1: Scenario Step 1 - Student Reports a Lost Item

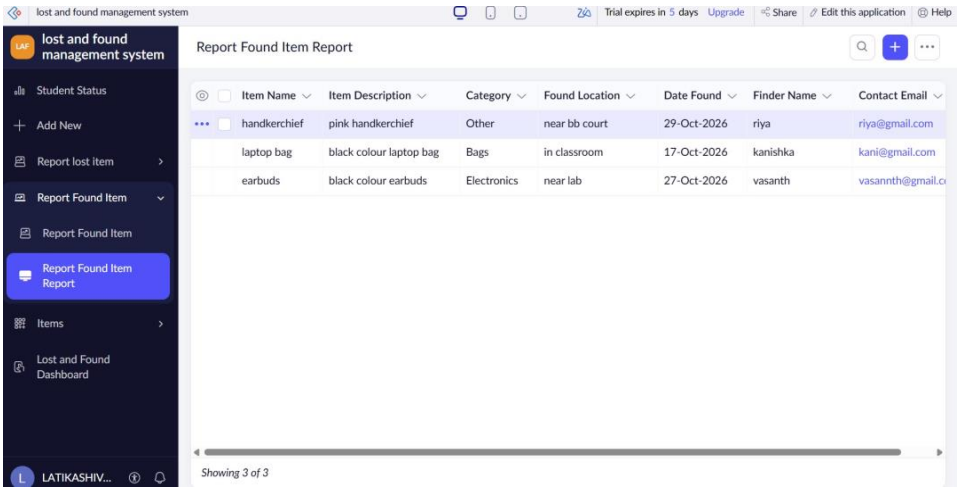


Figure 8.2: Scenario Step 2 - Finder Reports the Found Item

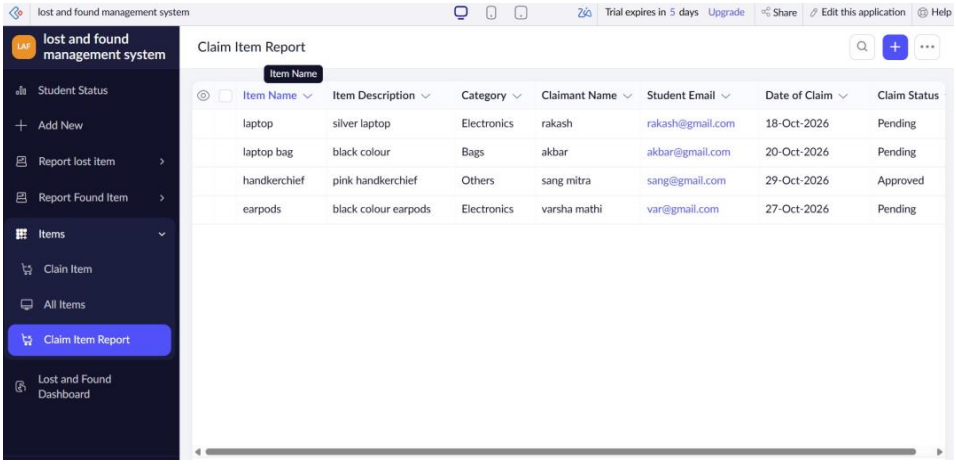


Figure 8.3: Scenario Step 3 - Claim Recorded and Approved

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented In	Outcome / Evidence
1. Forms and data model	Section 4.2, 5.2	Three forms with six field types (Figure 5.3, Figure 5.4, Figure 5.5)
2. Validation and notifications	Section 5.5, 5.6	Scripts and form workflows (Table 5.2, Table 5.3)
3. Reports and dashboard	Section 5.3, 5.4	Four list reports and one dashboard (Table 5.1, Figure 5.9)
4. Claim status tracking	Section 5.2, 5.3	Claim Status shown in the Claim Item Report (Figure 5.8)
5. Secure deployment and governance	Chapter 7	Roles, permissions, audit trail (Table 7.3)

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Record keeping	Paper registers in several offices	One shared, searchable record in three forms	Single source of truth
Item description	Free, unstructured	Standard fields: item name, category, location, date	Easier comparison
Search	Manual, from memory	Report search box and column sorting	Faster lookup
Claim record	Verbal	Claim Item form with date and Claim Status	Traceable handovers
Access point	Visit several desks	Dashboard with one button per task	One place for every task
Management view	None	Report record counts	Basic visibility

8.4 Limitations of the Low-Code Approach

Low-code has limits that were met in this project [5]. The first version does not match lost and found items automatically: staff compare the reports by eye, and a photograph-based or text-based match would need Deluge scoring or a pro-code

image service. Layouts are restricted to what the form and page builders allow. The application depends on one vendor's platform, and the trial or Education edition may limit the number of users, records or environments. Very large volumes of records would need performance care, such as indexed criteria and archiving.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of the application sits at the **no-code** end: forms, reports and the dashboard were built by dragging fields and setting properties. The **low-code** part is the Deluge scripting for validation and email notifications. No **pro-code** component was needed, but the design leaves room for one, as explained in Section 5.10.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Lost And Found Management System shows that a working campus application can be designed, built and tested on Zoho Creator by one developer in a short time. It replaces scattered paper registers with a single record of lost items, found items and claims, gives each a list report, records the claim status, and offers a dashboard from which every task can be started. The project applied forms, reports, a page, Deluge scripting, workflows and role-based access, and evidence for each is traced in Appendix A.

9.2 Future Enhancements

- Lookup fields to Category and Location master forms, so that values are consistent and can be analysed.
- Automatic matching of lost and found items with a Deluge scoring function and an Item Match form, with an email to the owner.
- Photograph upload on the report forms, with Zia Object Detection to suggest tags.
- A Handover form with a Signature field to record who collected an item.
- A Kanban report of claims by status, a calendar of found items, and charts of items by category on the dashboard.
- An approval process for claims, a scheduled reminder for items unclaimed for 30 days, and alerts to the security desk on Zoho Cliq.

9.3 Learning Outcomes and Career Relevance

The project gave practical experience in data modelling with forms, report design, Deluge scripting, workflow set-up, role-based access and the use of a trial environment. These skills map directly to the roles of citizen developer and low-code developer, and the limits met in Section 8.4 show the path from low-code to professional development.

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APPENDIX A - ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Each feature is marked as implemented and traced to the application component and to the report section that evidences it. Entries marked Y or N follow the screenshots of the application; entries marked [Y / N] in yellow cannot be seen in the screenshots and must be confirmed against the live application before the demonstration.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Menu with 6 entries; Design view	5.1	Figure 5.1
II	Forms with ≥ 8 distinct field types	N	6 types used: Name, Single Line, Multi Line, Drop Down, Date, Email	5.2	Figure 5.2
II	Lookup field (relationship between forms)	N	Not used; Category is a Drop Down	4.2, 5.2	-
II	Subform (parent-child data)	N	Not used	4.2, 5.2	-
II	Field validations / layout rules	Y	Email and Date field types; mandatory flags	5.2	Figure 5.3
II	List report with filter, grouping, custom action	Y	4 list reports (search, sort); custom action [Y]	5.3	Figure 5.6
II	Kanban / Calendar / Timeline / Map report	N	Not used	5.3	-
II	Pivot table / chart report	N	Not used	5.3	-
II	Page / dashboard built with UI Builder	Y	Lost and Found Dashboard	5.4	Figure 5.9
II	Deluge - form events (validate / success / field actions)	Y	DS-01 (confirmed) to DS-05	5.5	C.1 - C.5
II	Deluge - functions, fetch, loops, collections	N	Not used	5.5	-
II	Email / in-app notifications	Y	sendmail in DS-04, DS-05	5.5, 5.6	C.4, C.5
II	Form workflow	Y	5 form workflows (1 confirmed)	5.6	Figure 5.10
II	Scheduled workflow	N	Not used	5.6	-
II	Approval process	N	Not used (Claim_Status set manually)	5.6	-
II	Blueprint	N	Not used	5.6	-
II	Batch / report workflow	N	Not used	5.6	-
II	Testing & debugging evidence	Y	Test cases; info output	6	-
III	Integration with a Zoho service	Y	Zoho Mail	5.7	C.4
III	Connection / REST API call (invokeurl)	N	Not used	5.7	-

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
III	Webhook or Creator REST API	Y	Creator REST API tested in Postman	5.7	-
III	Users, roles and permission profiles	Y	Administrator, Student / Finder	5.8	-
III	Record-level access criteria / customer portal	Y	Public portal with 1 portal user	5.8, 5.11	Figure 5.11
III	Zia / AI feature	N	Not used (future enhancement)	5.9	-
III	Analytics (charts / Zoho Analytics)	N	Not used (future enhancement)	5.9	-
IV	Performance optimisation applied & measured	Y	Reduced columns; fetch criteria	7.2	Table 7.2
IV	Deployment across environments	Y	Stage and Production; version 1.0 in Production	7.1	Figure 7.1
IV	Security controls (encryption, OAuth, permissions)	Y	Permissions, OAuth token	7.3	Table 7.3
IV	Governance (audit trail, administration)	Y	Audit trail; settings	7.4	-
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Not used - justified in Section 5.10	5.10	-
V	Low-code limitations & spectrum analysis	Y	Sections 8.4 and 8.5	8.4, 8.5	-
-	Mobile app / portal access	Y	Portal created; Zoho Creator app	5.11	Figure 5.11

Total features implemented: 28 / 32

APPENDIX B - COURSE OUTCOME MAPPING

The self-assessed level uses 1 - Low, 2 - Medium, 3 - High; the examiner may revise it.

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	3	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1-5.6), 6	3	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7-5.9	3	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	2	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	2	

APPENDIX C - COMPLETE DELUGE SOURCE CODE

Every Deluge script of the application is given below, grouped by form or function. Script IDs correspond to Table 5.2.

C.1 Claim Item - On Successful Form Submission

```
// Script ID : DS-01   (confirmed in the application)
// Location  : Claim Item -> Workflow "Set Claim Status to Pending"
//             Created -> Successful form submission -> Execute without
condition
// Purpose   : Every new claim starts with the status Pending
input.Claim_Status = "Pending";
```

C.2 Report lost item - On Validate

```
// Script ID : DS-02
// Location  : Report lost item -> Form Events -> On Validate
// Purpose   : Reject a date of loss in the future
if(input.Date_Lost > zoho.currentdate)
{
        alert "Date of loss cannot be a future date.";
        cancel submit;
}
```

C.3 Report Found Item - On Validate

```
// Script ID : DS-03
// Location  : Report Found Item -> Form Events -> On Validate
// Purpose   : Reject a date found in the future
if(input.Date_Found > zoho.currentdate)
{
        alert "Date found cannot be a future date.";
        cancel submit;
}
```

C.4 Claim Item - On Success

```
// Script ID : DS-04
// Location  : Claim Item -> Form Events -> On Success
// Purpose   : Email the claimant that the claim was received
sendmail
[
        from : zoho.adminuserid
        to : input.Student_Email
        subject : "Claim received for " + input.Item_Name
        message : "Dear " + input.Claimant_Name.first_name + ",<br>Your
claim for <b>" + input.Item_Name + "</b> was recorded on " +
input.Date_of_Claim + " with status " + input.Claim_Status + ". The
administrator will verify it."
]
```

C.5 Claim Item - On Edit

```
// Script ID : DS-05
// Location  : Claim Item -> Form Events -> On Edit -> On Success
//             (workflow condition: Claim_Status is updated)
// Purpose   : Email the claimant when the claim is approved
if(input.Claim_Status == "Approved")
{
        sendmail
        [
                from : zoho.adminuserid
                to : input.Student_Email
                subject : "Your claim for " + input.Item_Name + " is
approved"
                message : "Dear " + input.Claimant_Name.first_name + ",
<br>Your claim has been approved. Please collect the item from the
administrator with your college ID."
        ]
}
```

APPENDIX D - USER MANUAL

D.1 Accessing the Application

Open the application URL in a web browser and sign in with the Zoho account that was invited. On a phone, install the Zoho Creator app, sign in with the same account and choose Lost And Found Management System.

D.2 Instructions for a Student / Finder

- **To report a lost item:** open Report lost item → Report lost item, enter your name, a description, the item name, the category, where you last saw it, the date and your email, and press Submit.
- **To report a found item:** open Report Found Item → Report Found Item, fill in the item details, the place and date found, your name and email, and press Submit.
- **To look for an item:** open the Report Found Item Report (or the dashboard button Found Items Report) and use the search box.
- **To claim an item:** open Items → Claim Item, enter the item details, your name and email and the date, and press Claim. You will be told when the claim is approved.

D.3 Instructions for the Administrator

- **To review reports:** use the dashboard buttons Lost Items Report, Found Items Report and Claims Report.
- **To decide a claim:** open Items → Claim Item Report, compare the claim with the found item, edit the record and set Claim Status to Approved.
- **To manage users:** invite users and assign a role in the application settings.
- **To manage categories:** edit the Category Drop Down values in the form builder.

APPENDIX E - EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	LATIKA SHIVANI N	**Register No.**	240701287
Project Title	Lost And Found Management System	**Date**	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90-100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70-89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50-69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner **Signature of External Examiner**

APPENDIX F - VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	LATIKA SHIVANI N	**Register No.**	240701287
Department / Section	Computer Science and Engineering	**Date of Viva**	
Application URL Verified	Yes / No	**Demo Duration**	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E - Excellent G - Good S - Satisfactory NI - Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1-8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER
Name: _____

EXTERNAL EXAMINER
Name: _____